



Reproductive Health

3.0

Reproductive Health

You have learnt about the **human reproductive system and its functions in Chapter 2. Reproductive health** is the closely related next idea - what it means, why it matters, and how it is achieved.

The term **simply refers to healthy reproductive organs with normal functions (F006)**, but it **has a broader perspective and includes the emotional and social aspects of reproduction also (F007)**.

- According to the **World Health Organisation (WHO), reproductive health means a total well-being in all aspects of reproduction, i.e., physical, emotional, behavioural and social (F008)**. Therefore, a society with people having physically and functionally normal reproductive organs and normal emotional and behavioural interactions among them in all sex-related aspects might be called **reproductively healthy (F009)**.

3.1

Reproductive Health - Problems and Strategies

India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal (F012). The summary states the same fact more strongly as **the first nation in the world to do so - both wordings are carried here (F012 SUMMARY-UNIQUE fold)**. These programmes called **family planning were initiated in 1951** and were **periodically assessed over the past decades (F013)**.

Improved programmes covering wider reproduction-related areas are currently in operation under the popular name Reproductive and Child Health Care (RCH) programmes (F014). Creating awareness among people about various reproduction related aspects and providing facilities and support for building up a **reproductively healthy society are the major tasks under these programmes (F015)**.

With the help of audio-visual and the print-media, governmental and non-governmental agencies have taken various steps to create awareness among the people about reproduction-related aspects (F016). Parents, other close relatives, teachers and friends, also have a major role in the dissemination of the above information (F017). **Introduction of sex education in schools should also be encouraged to provide right information to the young so as to discourage children from believing in myths and having misconceptions about sex-related aspects (F018)**.

Proper information about reproductive organs, adolescence and related changes, safe and hygienic sexual practices, sexually transmitted diseases (STD), AIDS, etc., would help people, especially those in the adolescent age group to lead a reproductively healthy life (F019).

Educating people, especially fertile couples and those in marriageable age group, about available birth control options, care of pregnant mothers, post-natal care of the mother and child, importance of breast feeding, equal opportunities for the male and the female child, etc., would address the importance of bringing up socially conscious healthy families of desired size (F020).

Awareness of problems due to uncontrolled population growth, social evils like sex-abuse and sex-related crimes, etc., need to be created to enable people to think and take up necessary steps to prevent them and thereby build up a socially responsible and healthy society (F021).

Successful implementation of various action plans to attain reproductive health requires strong infrastructural facilities, professional expertise and material support (F022). These are essential to provide medical assistance and care to people in reproduction-related problems like pregnancy, delivery, STDs, abortions, contraception, menstrual problems, infertility, etc. (F023). **Implementation of better techniques and new strategies from time to time are also required to provide more efficient care and assistance to people (F024)**.

Statutory ban on amniocentesis for sex-determination to legally check increasing menace of female foeticides, massive child immunisation, etc., are some programmes that merit mention in this connection (F025). NCERT prints the word as **aminocentesis**; the correct spelling is amniocentesis.

- **Amniocentesis:** some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances (F026). This procedure is used to test for the presence of certain genetic disorders such as, down syndrome, haemophilia, sickle-cell anemia, etc., determine the survivability of the foetus (F027) - the source sentence is garbled and prints haemoplilia for haemophilia, reproduced here with the correct term.

Research on various reproduction-related areas are encouraged and supported by governmental and non-governmental agencies to find out new methods and/or to improve upon the existing ones (F028).

① [NOTE] *Saheli - a new oral contraceptive for the females - was developed by scientists at Central Drug Research Institute (CDRI) in Lucknow, India (F029).*

Better awareness about sex related matters, increased number of medically assisted deliveries and better post-natal care leading to decreased maternal and infant mortality rates, increased number of couples with small families, better detection and cure of STDs and overall increased medical facilities for all sex-related problems, etc. all indicate improved reproductive health of the society (F030). The summary adds one more indicator to this list - assistance to infertile couples (F030 SUMMARY-UNIQUE fold).

3.2 Population Stabilisation and Birth Control

In the last century an all-round development in various fields significantly improved the quality of life of the people (F032). However, increased health facilities along with better living conditions had an explosive impact on the growth of population (F033) - this explosive growth is the population explosion of Exercise Q5 and the summary.

The world population which was around 2 billion (2000 million) in 1900 rocketed to about 6 billion by 2000 and 7.2 billion in 2011 (F034). A similar trend was observed in India too (F035, folded) - our population which was approximately 350 million at the time of our independence reached close to the billion mark by 2000 and crossed 1.2 billion in May 2011 (F036).

A rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR) as well as an increase in number of people in reproductive age are probable reasons for this (F037). Through our Reproductive Child Health (RCH) programme, though we could bring down the population growth rate, it was only marginal (F038). According to the 2011 census report, the population growth rate was less than 2 per cent, i.e., 20/1000/year, a rate at which our population could increase rapidly (F039). Such an alarming growth rate could lead to an absolute scarcity of even the basic requirements, i.e., food, shelter and clothing, in spite of significant progress made in those areas (F040). Therefore, the government was forced to take up serious measures to check this population growth rate (F041, folded) - the most important step to overcome this problem is to motivate smaller families by using various contraceptive methods (F042).

Advertisements in the media as well as posters/bills, etc., showing a happy couple with two children with a slogan Hum Do Hamare Do (we two, our two) (F043) illustrate the message. Many couples, mostly the young, urban, working ones have even adopted an one child norm (F044). Statutory raising of marriageable age of the female to 18 years and that of males to 21 years, and incentives given to couples with small families are two of the other measures taken to tackle this problem (F045).

- An ideal contraceptive should be user-friendly, easily available, effective and reversible with no or least side-effects (F047). It also should in no way interfere with the sexual drive, desire and/or the sexual act of the user (F048).

A wide range of contraceptive methods are presently available which could be broadly grouped into the following categories, namely Natural/Traditional, Barrier, IUDs, Oral contraceptives, Injectables, Implants and Surgical methods (F049). The table below compares them; the following paragraphs give the details named in it.

Category	Principle / How it works	Key examples / Notes
Natural / Traditional (F050)	Avoid chances of ovum and sperms meeting; no medicines or devices; side effects almost nil (F058) but chances of failure are also high (F059)	Periodic abstinence: avoid coitus day 10 to 17 of cycle (F051); this is the fertile period (F052). Withdrawal / coitus interruptus: male withdraws penis before ejaculation to avoid insemination (F054). Lactational amenorrhea: absence of menstruation; ovulation does not occur during intense lactation after parturition (F055); as long as mother breast-feeds fully, chances of conception are almost nil (F056), but effective only upto six months after parturition (F057)
Barrier (F060)	Ovum and sperms prevented from physically meeting with help of barriers; available for both males and females (F061)	Condoms (Fig. 3.1a, b): barriers made of thin rubber/latex sheath covering penis or vagina and cervix before coitus so ejaculated semen does not enter female tract (F062); Nirodh is a popular male brand (F064); protects also from STIs and AIDS (F065); disposable, self-inserted, gives privacy (F066). Diaphragms, cervical caps and vaults: rubber barriers inserted to cover cervix (F067); block entry of sperms through cervix and are reusable (F068). Spermicidal creams, jellies and foams usually used alongwith barriers increase efficiency (F069)
IUDs (F070-F076)	Inserted by doctors or expert nurses in uterus through vagina (F071); increase phagocytosis of sperms and Cu ions suppress sperm motility and fertilising capacity (F073); hormone-releasing IUDs in addition make uterus unsuitable for implantation and cervix hostile to sperms (F074); ideal for females who want to delay pregnancy and/or space children (F075); one of most widely accepted methods in India (F076)	Non-medicated: Lippes loop; Copper-releasing: CuT, Cu7, Multiload 375; Hormone-releasing: Progestasert, LNG-20 (Fig. 3.2) (F072)
Oral contraceptives (pills) (F077-F084)	Inhibit ovulation and implantation and alter quality of cervical mucus to prevent / retard entry of sperms (F081); very effective with lesser side effects and well accepted by females (F082)	Progestogens or progestogen-estrogen combinations taken as tablets, popularly called pills (F077-F078); taken daily for 21 days starting preferably within first 5 days of menstrual cycle; after a gap of 7 days (menstruation occurs) repeat same pattern till desire to prevent conception (F079-F080). Saheli: contains non-steroidal preparation; once a week pill with very few side effects and high contraceptive value (F083-F084)
Injectables / Implants (F085-F086)	Mode of action similar to pills; effective periods much longer (F086)	Progestogens alone or in combination with estrogen can also be used as injections or implants under skin (Fig. 3.3) (F085)
Emergency contraception (F087)	Within 72 hours of coitus , very effective to avoid possible pregnancy due to rape or casual unprotected intercourse	Progestogens or progestogen-estrogen combinations or IUDs within 72 hours of coitus (F087)
Surgical (sterilisation) (F088-F093)	Blocks gamete transport and thereby prevents conception (F089); highly effective but reversibility very poor (F093)	Generally advised as terminal method to prevent any more pregnancies (F088). Male - vasectomy and female - tubectomy (F090). Vasectomy: small part of vas deferens is removed or tied up through small incision on scrotum (Fig. 3.4a) - Vas deferens tied and cut (F091). Tubectomy: small part of fallopian tube is removed (Fig. 3.4b) or tied up through incision in abdomen or through vagina - Fallopian tubes tied and cut (F092)

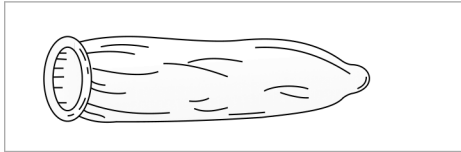


Fig. 3.1a - Condom for the male. A barrier made of thin rubber/latex sheath used to cover the penis so that ejaculated semen does not enter the female reproductive tract.

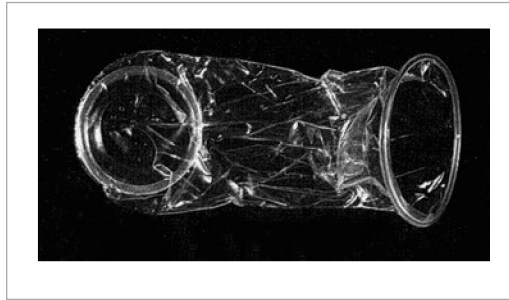


Fig. 3.1b - Condom for the female. A barrier made of thin rubber/latex sheath used to cover the vagina and cervix so that ejaculated semen does not enter the female reproductive tract.

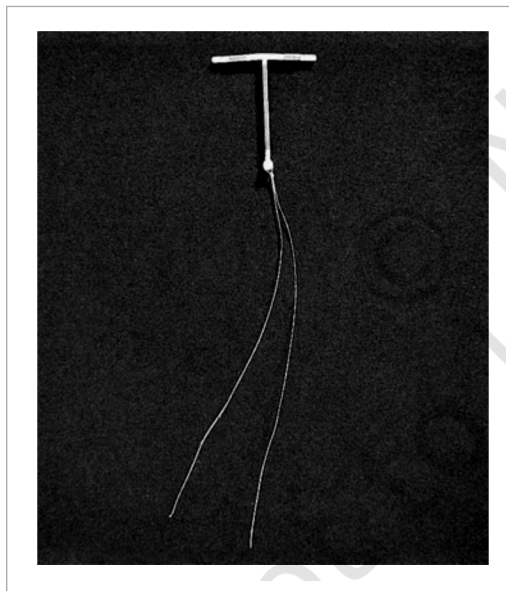


Fig. 3.2 - Copper T (CuT). An example of a copper-releasing IUD inserted in the uterus through the vagina.

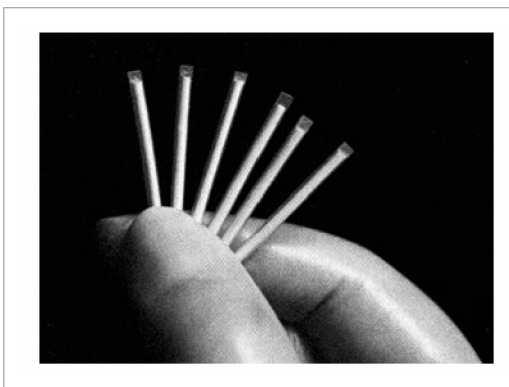


Fig. 3.3 - Implants. Progestogen or progestogen-estrogen implants placed under the skin; mode similar to pills but effective for much longer.

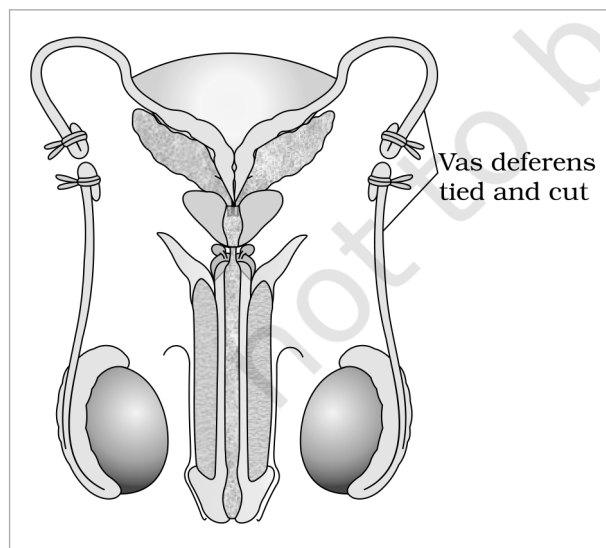


Fig. 3.4a - Vasectomy. A small part of the vas deferens is removed or tied up through a small incision on the scrotum. Label: Vas deferens tied and cut.

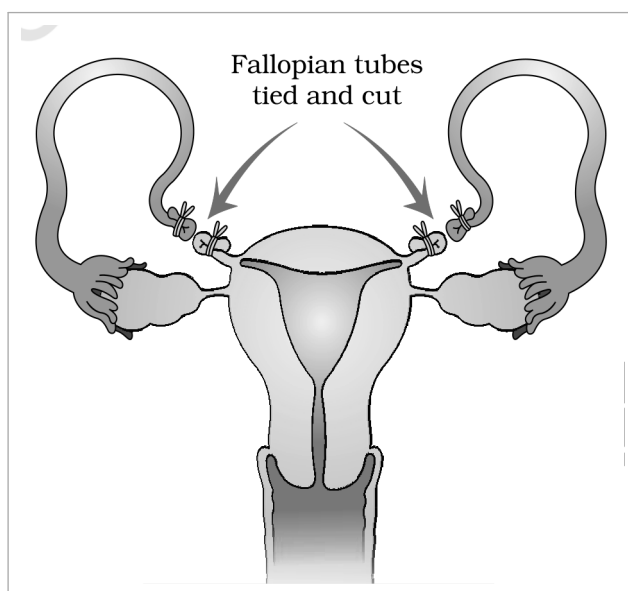


Fig. 3.4b - Tubectomy. A small part of the fallopian tube is removed or tied up through an incision in the abdomen or through the vagina. Label: Fallopian tubes tied and cut.

Selection of a suitable contraceptive method and its use should always be undertaken in consultation with qualified medical professionals (F094).

Contraceptives are not regular requirements for the maintenance of reproductive health (F095). In fact, they are practiced against a natural reproductive event, i.e., conception/pregnancy (F096). One is forced to use these methods either to prevent pregnancy or to delay or space pregnancy due to personal reasons (F097).

No doubt, the widespread use of these methods have a significant role in checking uncontrolled growth of population (F098). However, their possible ill-effects like nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding or even breast cancer, though not very significant, should not be totally ignored (F099).

☆ **[MEMORY AID - not in NCERT]** To recall the seven contraceptive categories in NCERT order, remember **N-B-I-O-I-I-S**: **N**atural, **B**arrier, **I**UDs, **O**ral (pills), **I**njectables, **I**mplants, **S**urgical. Pills need **21 + 7** days; emergency is **72 hours**; lactational cover is **6 months** only.

- **Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy (MTP) or induced abortion (F101) - the opener defines the term used in its own section heading.**

Nearly 45 to 50 million MTPs are performed in a year all over the world which accounts to 1/5th of the total number of conceived pregnancies in a year (F102). Whether to accept / legalise MTP or not is being debated upon in many countries due to emotional, ethical, religious and social issues involved in it (F103).

Government of India legalised MTP in 1971 with some strict conditions to avoid its misuse (F104). Such restrictions are all the more important to check indiscriminate and illegal female foeticides which are reported to be high in India (F105).

Why MTP? Obviously the answer is - to get rid of unwanted pregnancies either due to casual unprotected intercourse or failure of the contraceptive used during coitus or rapes (F106, as a question-and-answer pair). MTPs are also essential in certain cases where continuation of the pregnancy could be harmful or even fatal either to the mother or to the foetus or both (F107).

MTPs are considered relatively safe during the first trimester, i.e., upto 12 weeks of pregnancy (F108). Second trimester abortions are much more riskier (F109).

One disturbing trend observed is that a majority of the MTPs are performed illegally by unqualified quacks which are not only unsafe but could be fatal too (F110). Another dangerous trend is the misuse of amniocentesis to determine the sex of the unborn child (F111). Frequently, if the foetus is found to be female, it is followed by MTP - this is totally against what is legal (F112) and such practices should be avoided because these are dangerous both for the young mother and the foetus (F113, folded onto F112).

Effective counselling on the need to avoid unprotected coitus and the risk factors involved in illegal abortions as well as providing more health care facilities could reverse the mentioned unhealthy trend (F114).

① [NOTE] The Medical Termination of Pregnancy (Amendment) Act, 2017 was enacted by the government of India with the intension of reducing the incidence of illegal abortion and consequent maternal mortality and morbidity (F115) - source prints intension for intention.

Under the Act, a pregnancy may be terminated on certain considered grounds and within strict time limits:

- 1 Within the first 12 weeks of pregnancy, on the opinion of **one registered medical practitioner** (F116).
- 2 More than 12 weeks but fewer than 24 weeks, on the opinion of **two registered medical practitioners**, formed in good faith, that the required grounds exist (F117).
- 3 Ground (i): **continuation of the pregnancy would involve a risk to the life of the pregnant woman or of grave injury to her physical or mental health** (F118).
- 4 Ground (ii): **there is a substantial risk that if the child were born, it would suffer from such physical or mental abnormalities as to be seriously handicapped** (F118).

3.4 Sexually Transmitted Infections (STIs)

- Infections or diseases which are transmitted through sexual intercourse are collectively called **sexually transmitted infections (STI) or venereal diseases (VD) or reproductive tract infections (RTI)** (F120) - the opener defines the term used in its own heading.

Gonorrhoea, syphilis, genital herpes, chlamydiasis, genital warts, trichomoniasis, hepatitis-B and of course, the most discussed infection in the recent years, HIV leading to AIDS are some of the common STIs (F121). Among these, HIV infection is most dangerous and is discussed in detail in Chapter 7 (F122).

Some of these infections like hepatitis-B and HIV can also be transmitted by sharing of injection needles, surgical instruments, etc., with infected persons, transfusion of blood, or from an infected mother to the foetus too (F123).

Except for hepatitis-B, genital herpes and HIV infections, other diseases are completely curable if detected early and treated properly (F124).

Early symptoms of most of these are minor and include itching, fluid discharge, slight pain, swellings, etc., in the genital region (F125). Infected females may often be asymptomatic and hence, may remain undetected for long (F126). Absence or less significant symptoms in the early stages of infection and the social stigma attached to the STIs, deter the infected persons from going for timely detection and proper treatment (F127).

This could lead to complications later, which include pelvic inflammatory diseases (PID), abortions, still births, ectopic pregnancies, infertility or even cancer of the reproductive tract (F128). STIs are a major threat to a healthy society (F129). Therefore, prevention or early detection and cure of these diseases are given prime consideration under the reproductive health-care programmes (F130).

Though all persons are vulnerable to these infections, their incidences are reported to be very high among persons in the age group of 15-24 years - the age group to which you also belong (F131).

There is no reason to panic because prevention is possible (F132, folded) - three simple principles keep one free of these infections:

- 1** Avoid sex with unknown partners/multiple partners (F133 i).
- 2** Always try to use condoms during coitus (F133 ii).
- 3** In case of doubt, go to a qualified doctor for early detection and get complete treatment if diagnosed with infection (F133 iii).

3.5 Infertility

A discussion on reproductive health is incomplete without a mention of infertility (F135, folded). A large number of couples all over the world including India are infertile, i.e., they are unable to produce children inspite of unprotected sexual co-habitation (F136) - the summary adds the diagnostic criterion even after 2 years of unprotected sexual cohabitation, which is carried here (F136 SUMMARY-UNIQUE fold).

The reasons for this could be many - physical, congenital, diseases, drugs, immunological or even psychological (F137). In India, often the female is blamed for the couple being childless, but more often than not, the problem lies in the male partner (F138).

Specialised health care units (infertility clinics, etc.) could help in diagnosis and corrective treatment of some of these disorders and enable these couples to have children (F139). However, where such corrections are not possible, the couples could be assisted to have children through certain special techniques commonly known as assisted reproductive technologies (ART) (F140).

- In vitro fertilisation (IVF - fertilisation outside the body in almost similar conditions as that in the body) followed by embryo transfer (ET) is one of such methods (F141).

In this method, popularly known as test tube baby programme, ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form zygote under simulated conditions in the laboratory (F142).

The zygote or early embryos (with upto 8 blastomeres) could then be transferred into the fallopian tube (ZIFT - zygote intra fallopian transfer) and embryos with more than 8 blastomeres, into the uterus (IUT - intra uterine transfer), to complete its further development (F143). Embryos formed by in-vivo fertilisation (fusion of gametes within the female) also could be used for such transfer to assist those females who cannot conceive (F144).

- Transfer of an ovum collected from a donor into the fallopian tube (GIFT - gamete intra fallopian transfer) of another female who cannot produce one, but can provide suitable environment for fertilisation and further development is another method attempted (F145).
- Intra cytoplasmic sperm injection (ICSI) is another specialised procedure to form an embryo in the laboratory in which a sperm is directly injected into the ovum (F146).

Infertility cases either due to inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates, could be corrected by artificial insemination (AI) technique (F147). In this technique, the semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus (IUI - intra-uterine insemination) of the female (F148).

Though options are many, all these techniques require extremely high precision handling by specialised professionals and expensive instrumentation (F149). Therefore, these facilities are presently available only in very few centres in the country (F150) and obviously their benefits is affordable to only a limited number of people (F151, folded with F150).

Emotional, religious and social factors are also deterrents in the adoption of these methods (F152).

Since the ultimate aim of all these procedures is to have children, in India we have so many orphaned and destitute children, who would probably not survive till maturity, unless taken care of (F153). Our laws permit legal adoption and it is as yet, one of the best methods for couples looking for parenthood (F154).

① [NOTE] ART, GIFT and IUI are not interchangeable - IVF-ET/ZIFT/IUT use fertilisation outside the body followed by embryo transfer; GIFT transfers the unfertilised ovum; ICSI injects one sperm into one ovum; AI/IUI introduces semen into the uterus/vagina - and the blastomere rule decides ZIFT (upto 8) versus IUT (more than 8).

QR

Quick Recap

- **Reproductive health** is total well-being in all aspects - physical, emotional, behavioural and social (WHO, F008); it needs counselling, awareness about organs, adolescence, safe hygiene, STIs including AIDS, plus medical care for menstrual problems, pregnancy, delivery, MTP, STIs, birth control, infertility and post-natal management (F015/F023); India was amongst the first / the first nation to launch RCH programmes (1951 family planning) (F012-F014), now judged by reduced MMR/IMR, more small families, better STD cure, more medical facilities and assistance to infertile couples (F030).
- **Population explosion:** better health and living conditions drove world 2B (1900) to 6B (2000) to 7.2B (2011) (F034) and India 350M at independence to near 1B by 2000 to over 1.2B in May 2011 (F035-F036); reasons fall in death/MMR/IMR and more people in reproductive age (F037); RCH cut growth only marginally to less than 2% = 20/1000/year (2011) (F038-F039), threatening food, shelter, clothing scarcity (F040) and prompting smaller-family motivation, Hum Do Hamare Do, one-child norm, marriage age 18F/21M and incentives (F042-F045).
- **Contraception** - seven NCERT groups: an ideal contraceptive is user-friendly, easily available, effective, reversible, with least side effects and no interference with drive/act (F047-F048); Natural methods avoid ovum-sperm meeting without devices - periodic abstinence day 10-17 fertile period, withdrawal, lactational amenorrhea (intense lactation, 6 months only, almost nil side effects but high failure) (F050-F059); Barrier methods physically block meeting - condoms (Nirodh, protects also from STIs/AIDS, disposable, private), diaphragms/caps/vaults (reusable), plus spermicidal creams/jellies/foams (F060-F069); IUDs inserted by doctors/nurses through vagina - Lippes loop, CuT/Cu7/Multiload 375, Progestasert/LNG-20 via phagocytosis, Cu suppression, hormone hostile uterus/cervix; ideal for spacing, most widely accepted (F070-F076); Pills (progestogen or progestogen-estrogen) 21 days within first 5 days + 7-day gap, inhibit ovulation/implantation and alter cervical mucus, well accepted and Saheli weekly non-steroidal (F077-F084); Injectables/implants under skin, similar mode but much longer action (F085-F086); emergency within 72 hours (F087); Surgical sterilisation blocks gamete transport, vasectomy (Vas deferens tied and cut) and tubectomy (Fallopian tubes tied and cut), highly effective, poorly reversible (F088-F093); always choose with a qualified doctor (F094); contraception is against natural conception, used only to prevent/delay/space pregnancy, with possible but not very significant ill-effects (nausea, bleeding, breast cancer ignored at peril) (F095-F099).
- **MTP:** intentional/voluntary termination before full term, induced abortion (F101); 45-50 million per year = 1/5th of conceived pregnancies (F102); debated for emotional/ethical/religious/social reasons (F103); legal in India since 1971 with strict conditions to avoid misuse and illegal female foeticide (F104-F105); needed for unwanted pregnancies (casual/failed contraception/rape) or harm to mother/foetus (F106-F107); relatively safe upto 12 weeks, riskier in second trimester (F108-F109); majority illegal by quacks, unsafe/fatal; misuse of amniocentesis for sex determination leading to female foeticide, dangerous for mother and foetus, reversible only by counselling, avoiding unprotected coitus and more health facilities (F110-F114); Amendment Act 2017 aims to cut illegal abortion/mortality (F115); within 12 weeks - one doctor's opinion; 12-24 weeks - two doctors; grounds: risk to life/grave injury to physical/mental health, or substantial risk of serious physical/mental handicap in child (F116-F118).
- **STIs:** infections transmitted by sexual intercourse - STI/VD/RTI (F120): gonorrhoea, syphilis, genital herpes, chlamydia, genital warts, trichomoniasis, hepatitis-B, HIV-AIDS (most dangerous, Chapter 7) (F121-F122); hepatitis-B/HIV also spread by needles, surgical instruments, blood transfusion, mother-to-foetus (F123); all except hepatitis-B, genital herpes and HIV are completely curable if early and properly treated (F124); minor early symptoms - itching, discharge, pain, swelling; females often

asymptomatic (F125-F126); **absent/less symptoms plus stigma deter timely treatment** (F127) leading to **PID, abortions, still births, ectopic pregnancies, infertility, reproductive tract cancer** (F128); **major threat, prevention/early cure is prime under RCH** (F129-F130); **highest incidence 15-24 years (your age)** (F131); prevention is possible - **avoid unknown/multiple partners, use condoms, see a qualified doctor if in doubt and complete treatment** (F132-F133).

- **Infertility: inability to produce children inspite of unprotected co-habitation even after 2 years (summary criterion)** (F136); many causes - physical, congenital, diseases, drugs, immunological, psychological (F137); female often blamed but more often male partner is responsible (F138); infertility clinics help **diagnosis/corrective treatment**, but where not possible, assisted reproductive technologies (ART) assist (F139-F140); **IVF-ET (test tube baby): ova and sperms collected, zygote formed in lab under simulated conditions; ZIFT (upto 8 blastomeres) to fallopian tube, IUT (more than 8) to uterus; in-vivo fertilised embryos also usable** (F141-F144); **GIFT transfers donor ovum to another female's fallopian tube who provides environment** (F145); **ICSI injects one sperm directly into ovum** (F146); **AI corrects inability to inseminate or very low sperm counts - semen (husband or donor) artificially introduced into vagina or uterus (IUI)** (F147-F148); all require high precision, expensive instrumentation, available in very few centres, affordable to limited people, deterred by emotional/religious/social factors (F149-F152); **India has many orphaned/destitute children unlikely to survive without care - legal adoption is as yet one of the best methods for parenthood** (F153-F154).

EX

Terms Used in the Exercises

Only three exercise parts rely on ideas the chapter never states. Each is answered below using only chapter-based reasoning; every other exercise (Q1-Q6, Q8-Q10, Q11b-d, Q12a-b,d) is answered by the body text above.

- **Q7 - Why removal of gonads cannot be a contraceptive option.** The chapter's surgical methods are **sterilisation by blocking gamete transport - Vas deferens tied and cut (vasectomy) and Fallopian tubes tied and cut (tubectomy)** (F089-F092); they **do not remove the gonads (testes/ovaries)**, and **reversibility is already very poor** (F093). Removing the testes or ovaries would **remove the source of gametes and of androgens/estrogens** (the hormones that maintain secondary sexual characters and reproductive function) - a permanent, non-contraceptive castration that the chapter never lists among its seven contraceptive categories (F049) and that violates the **ideal contraceptive criterion of reversibility and no interference with normal physiology** (F047). Hence gonad removal is not a contraceptive option.
- **Q11(a) - True: abortions can occur spontaneously.** The chapter defines only **intentional/voluntary termination (MTP/induced abortion)** (F101), but **spontaneous abortion (miscarriage) is a different fact** - the chapter itself lists **abortion as a complication of STIs/PID** (F128) that occurs without any MTP procedure, which shows that pregnancy loss can happen on its own.
- **Q12(c) - False to correct: Oral pills are not particularly popular among rural women.** The chapter states only that **pills are very effective with lesser side effects and are well accepted by the females** (F082) and that **Saheli is a once-a-week pill** (F084), but **never distinguishes rural from urban users**. In fact, **IUDs are described as one of the most widely accepted methods in India** (F076), often promoted for spacing in the public programme, so singling out pills as rural-popular is not supported by the text.

Every fact, number, name, qualifier, table row, figure and figure label in NCERT Class 12 Chapter 3 is carried above. Nothing outside the source chapter has been added, except the clearly marked NOTE/MEMORY AID material and the exercise-gap explanations, which are derived only from chapter content.